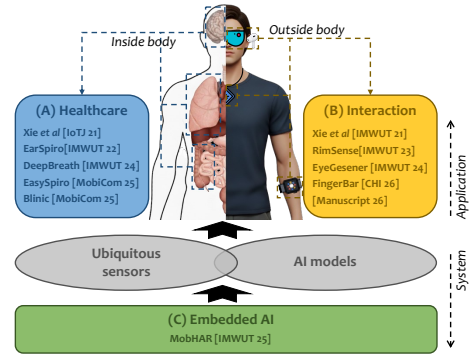


Wentao Xie

Research

My research tackles human-centric problems through **ubiquitous sensors** and **Artificial Intelligence of Things (AIoT)** systems. I design ubiquitous sensing platforms—spanning embedded devices, on-body and ambient sensors, and IoT infrastructures—and develop robust AI models that lower the barrier to measuring and interpreting human activities. These activities range from those happening inside our body (e.g., respiration, cognition) to more obvious physical activities (e.g., gestures, motions), enabling applications such as **home-based healthcare** and **HCI on edge devices**. My research is highly interdisciplinary and transferable, where I work closely with domain experts such as health professionals and UX/UI designers to identify physical world problems and solve them using ubiquitous sensing methods. Some of my research outputs are currently being deployed to benefit end users.



Keywords: Ubiquitous computing, HCI, AIoT, cyber-physical systems (CPS), smart healthcare, wearables

Appointment

Sep 2024– **The Hong Kong University of Science and Technology**, Hong Kong
Department of Computer Science and Engineering
Research Assistant Professor

Education

2019–2024 **The Hong Kong University of Science and Technology (HKUST)**, Hong Kong
Ph.D., Computer Science and Engineering
Thesis: *Understand Human Behaviours with IoT Sensors: From Physical to Physiological Sensing*
Supervisor: Prof Qian Zhang

2015–2019 **Southern University of Science and Technology (SUSTech)**, Shenzhen, China
B.Eng., Computer Science and Engineering

Achievements and Awards

Feb 2026 Gold Medal with jury congratulations, International Exhibition of Inventions Geneva
Oct 2023 Research Travel Grants, HKUST
Feb 2023 Research Travel Grants, HKUST
Aug 2022 HKUST RedBird Academic Excellence Award, HKUST
Oct 2021 HKTIIT Post-Graduate Excellence Scholarship, Hong Kong Telecom Institute of Information Technology (HKTIIT)
Aug 2019 Distinguished Final-year Thesis, SUSTech

Grants

Jun 2026 PI, **General Research Fund**, Hong Kong Research Grant Council

Teaching

- Spring 2025 **Instructor**, COMP4021 - *Internet Computing*, HKUST
Fall 2025 Students will learn full-stack web application programming skills, including web languages such as HTML, JavaScript, and PHP, and common libraries/frameworks such as jQuery, AJAX, Express, etc. Course modules contain lectures, labs, and projects.
Student feedback rating: 4.4/5, department average: 4.15, university average: 4.05
- Spring 2025 **Instructor**, UROP1100-2100 - *Undergraduate Research Opportunities Series*, HKUST
-Fall 2025 Lead a small group of undergraduate students in engaging with research activities, including idea brainstorming, prototype implementation, and empirical studies.
- Fall 2022 **Teaching Assistant**, COMP4621 - *Computer Communication Networks I*, HKUST
Delivered lab tutorials and designed the course project
- Fall 2022 **Teaching Assistant**, COMP4531 - *Mobile Computing and Smart Sensing*, HKUST
Prepared course lab materials and designed the course project.
- Fall 2022 **Teaching Assistant**, COMP4021 - *Internet Computing*, HKUST
Delivered lab tutorials and graded lab assignments.

Supervision and Mentoring

- 2026 **Jaime Salafranca Pardo**, *Undergraduate research student*, HKUST
- Fall 2025– **Yuxin Tan**, *Mentored PhD student*, HKUST
- Spring 2025– **Yixuan Liu**, *Undergraduate research intern*, HKUST
Fall 2025
- Spring 2025– **Lambo Qin**, *Undergraduate research intern*, HKUST
IMWUT'26 in submission
- Spring 2025– **Yankai Zhao**, *Visiting Master's student*, SUSTech
CHI'26
- Fall 2024– **Akshat Agarwal, Sukruti Rai**, *Final-year Project students*, HKUST
Spring 2025 Project title: *Stock Exchange Simulator*
- Fall 2024– **Yizhen Zhang**, *Mentored PhD student*, HKUST
UbiComp'26
- Fall 2023– **Chi Xu**, *Mentored PhD student*, HKUST
UbiComp'24, MobiCom'25 and one UbiComp'26 paper in submission

Professional Services

- Conference **ACM MobiCom Competition Co-chair**, 2025
Organizing
- Technical **ACM MobiHoc**, 2025, 2026
Program **EAI MobiQuitous**, 2026
Committee **ACM ISWC**, 2025, 2026
IEEE GLOBECOM, 2025
- Reviewer **ACM IMWUT** (2 Recognitions for Outstanding Reviews), 2022-2026
ACM CHI, 2025
ACM Transactions on Computing for Healthcare, 2025, 2026
IEEE Transactions on Mobile Computing, 2025, 2026
IEEE Transactions on Parallel and Distributed Systems, 2026
ACM Multimedia, 2025
ACM ISWC, 2023

Publications

(* denotes co-primary authors, ____ denotes students under my supervision/mentorship)

- [19] Yankai Zhao*, **Wentao Xie***, Haorui Li, Jiao Li, Tao Sun, Qian Zhang, and Jin Zhang, "FingerBar: A Mid-Air Touch Bar Interface for Earphones Using Finger-Generated Acoustics, " in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. (**CHI 2026**)
- [18] Yizhen Zhang, Jinjian Wang, **Wentao Xie**, Qingyong Hu, Haiyan Hu, Guihua Li, Qian Zhang "PIGDAssess: Wearable Dual-Task Sensing for Self-Administered PIGD Assessment in Parkinson's Disease," in *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* (**UbiComp 2026**)
- [17] Baichen Yang, Xinyi Zhang, Xin He, Chi Xu, **Wentao Xie**, Zuru Liang, Patrick Shu-hang Yung, and Qian Zhang, "ACLGard: Physics-Aware Knee Loading Monitoring System for Anterior Cruciate Ligament Injury Prevention Training, " to appear in *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 9, no. 4, 2025. (**UbiComp 2026**)
- [16] **Wentao Xie**, "Decoding Your Body with Ubiquitous Sensing - Wearable and Contactless Perception for Health and Beyond," in *Proceedings of the 2025 ACM CoNEXT Workshop Edge-Cloud Collaboration for AI, 2025*. (**CoNEXT 2025 Workshop**)
- [15] Chi Xu*, **Wentao Xie***, Baichen Yang, Yizhen Zhang, Yanbin Gong, Jin Zhang, Wei Li, Shifang Yang, and Qian Zhang, "EasySpiro: Assessing Lung Function via Arbitrary Exhalations on Commodity Earphones, " *Proceedings of the 31th Annual International Conference on Mobile Computing and Networking, Hong Kong, Nov. 2025*. (**MobiCom 2025**) **Being deployed to Taikang Senior Living. Gold Medal with jury congratulations @ Geneva Invention Expo**
- [14] Meng Xue*, **Wentao Xie***, Huizi Yizuo, Shumao Wu, Zhilong Zhang, Yinan Zhu, Qian Zhang, and Changzheng Chen, "Home-based Dry Eye Assessment via Blink Kinematics Using mmWave and Clinical Knowledge Distillation, " to appear in *Proceedings of the 31th Annual International Conference on Mobile Computing and Networking, Hong Kong, Nov. 2025*. (**MobiCom 2025**)
- [13] Yetong Cao, Dong Ma, **Wentao Xie**, Qian Zhang, and Jun Luo, "ESPIRO: Natural Pulmonary Function Monitoring via Earphone-Acquired Speech," *Proceedings of the 31th Annual International Conference on Mobile Computing and Networking (MobiCom '25), Hong Kong, Nov. 2025*. (**MobiCom 2025**)
- [12] Yanbin Gong, **Wentao Xie**, Chi Xu, Qian Zhang, and Shifang Yang, "SputumLocator: Enhancing Airway Clearance with Auscultation-Based Sputum Localization, " to appear in *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 9, no. 2, 2025. (**UbiComp 2025**)
- [11] Meng Xue, Yinan Zhu, **Wentao Xie**, Zhixian Wang, Yanjiao Chen, Kui Jiang, and Qian Zhang, "MobHAR: Source-free Knowledge Transfer for Human Activity Recognition on Mobile Devices, " *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 9, no. 1, 2025. (**UbiComp 2025**)
- [10] Tao Sun, Yankai Zhao, **Wentao Xie**, Jiao Li, Yongyu Ma, and Jin Zhang, "EyeGesener: Eye Gesture Listener for Smart Glasses Interaction Using Acoustic Sensing, " in *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 8, no. 3, 2024. (**UbiComp 2024**)
- [9] **Wentao Xie***, Chi Xu*, Yanbin Gong, Yu Wang, Yuxin Liu, Jin Zhang, Qian Zhang, Zeguang Zheng, and Shifang Yang, "DeepBreath: Breathing Exercise Assessment with a Depth Camera," in *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 7, no. 4, 2024. (**UbiComp 2024**) **Gold Medal with jury congratulations @ Geneva Invention Expo**
- [8] Yanbin Gong, **Wentao Xie**, Qian Zhang, and Shifang Yang, "Hypergradient Descent Based Multi-Task Learning on Auscultation Point Guided Respiratory Sound Classification," in *2024 IEEE 20th International Conference on Body Sensor Networks, Chicago IL, USA, Oct. 2024*. (**BSN 2024**)
- [7] Linfei Ge, **Wentao Xie**, Jin Zhang, Qian Zhang, "BLEAR: Practical Wireless Earphone Tracking under BLE Protocol", in *The Proceedings of 2023 IEEE International Conference on Pervasive Computing and Communications, Biarritz, France, Mar. 2024*. (**PerCom 2024**)

- [6] **Wentao Xie**, Huangxun Chen, Jing Wei, Jin Zhang, Qian Zhang, "RimSense: Enabling Touch-based Interaction on Eyeglass Rim using Piezoelectric Sensors", in *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 7, no. 4, 2023. (**UbiComp 2024**)
- [5] Baichen Yang, Qingyong Hu, **Wentao Xie**, Xinchun Wang, Wei Luo, Qian Zhang, "PDAssess: A Privacy-preserving Free-speech based Parkinson's Disease Daily Assessment System", in *Proceedings of the 21st ACM Conference on Embedded Networked Sensor Systems, Istanbul, Turkiye, Nov. 2023*. (**SenSys 2023**)
- [4] **Wentao Xie**, Qingyong Hu, Jin Zhang, Qian Zhang "EarSpiro: Earphone-based Spirometry for Lung Function Assessment" in *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 6, no. 4, pp. 1–27, 2022. (**UbiComp 2022**)
- [3] **Wentao Xie**, Jin Zhang, Qian Zhang "Poster: Transforming Eyeglass Rim into Touch Panel Using Piezoelectric Sensors" in *Proceedings of the 28th Annual International Conference on Mobile Computing and Networking, Sydney NSW Australia, Oct. 2022*. (**MobiCom 2022**)
- [2] **Wentao Xie**, Qian Zhang, Jin Zhang, "Acoustic-based Upper Facial Action Recognition for Smart Eyewear" in *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)*, vol. 5, no. 2, pp. 1–28, 2021. (**UbiComp 2021**)
- [1] **Wentao Xie**, Runxin Tian, Jin Zhang, Qian Zhang, "Noncontact Respiration Detection Leveraging Music and Broadcast Signals" in *IEEE Internet of Things Journal*, vol. 8, no. 4, pp. 2931-2942, 2021. (**IoTJ 2021**)

Manuscripts Under Review

- [2] **Wentao Xie**, Chi Xu, Lambo Qin, Yixuan Liu, Yankai Zhao, Qian Zhang "Swift: Turning Your Hand into a Writing Pad with Unmodified Smartwatches," submitted to *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* (**UbiComp 2026**)
- [1] Chi Xu, **Wentao Xie**, Huaning Tan, Zirui Huang, Yanbin Gong, Shifang Yang, Qian Zhang, "CaRe-Fit: Cost-Effective Cardiorespiratory Fitness Assessment Using Multimodal Sensors," submitted *The Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)* (**UbiComp 2026**)